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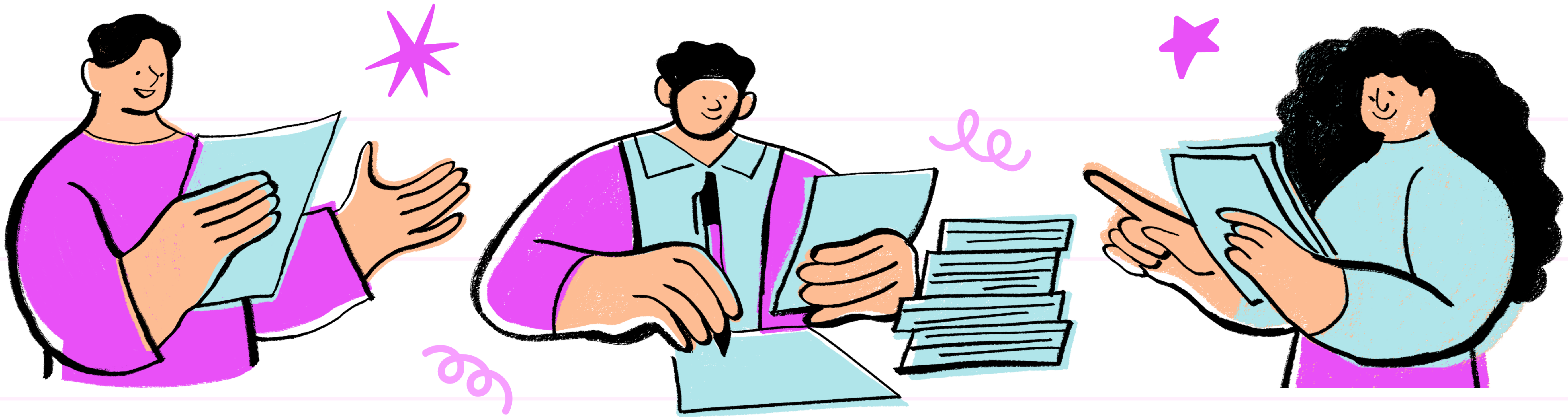
Machine Learning vs. Deep Learning vs. Generative AI

Understanding AI Basics



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Introduction

- Artificial Intelligence (AI) is a broad term that includes Machine Learning (ML), Deep Learning (DL), and Generative AI.

- These are different ways computers can learn and make decisions.

- This presentation will explain them in simple terms.

Machine Learning

- ML is a way for computers to learn from examples instead of being programmed for every task.
- Types of ML:
  - Supervised Learning: Learns from labeled examples (e.g., email spam filter).
  - Unsupervised Learning: Finds patterns in data without labels (e.g., customer segmentation).
  - Reinforcement Learning: Learns by trial and error (e.g., game-playing AI).
- Examples: Spam filters, product recommendations, fraud detection.

Generative AI

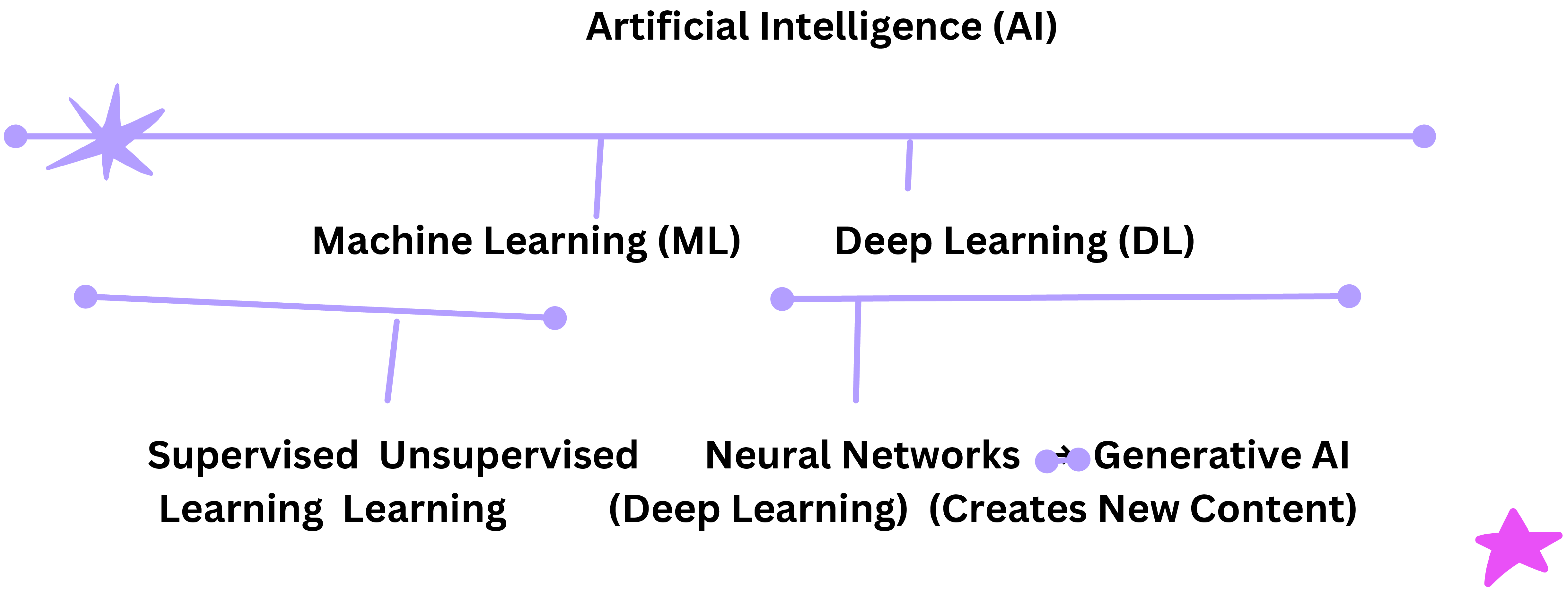
- Generative AI creates new things like images, text, and music based on patterns it has learned.
- How it Works: Uses special AI models like GANs (Generative Adversarial Networks) and Transformers.
- Examples: ChatGPT (text generation), DALL-E (AI art), AI-generated music and videos.

Deep Learning

- DL is a more advanced type of ML that mimics how the human brain works using artificial neural networks.
- Key Features:
- Uses layers of connected nodes (neurons) to process information.
- Needs a lot of data and computing power.
- Examples: Self-driving cars, voice assistants (Siri, Alexa), face recognition.

Key Differences

Feature	Machine Learning	Deep Learning	Generative AI
Definition	Learns from data	Uses neural networks	Creates new content
Complexity	Simple to moderate	High	Very High
Data Requirement	Moderate	Large	Very Large
Computing Power	Simple to moderate	High	Very High
Example Uses	Spam filters, recommendations	Self-driving cars, voice assistants	AI-generated text, art, videos


Artificial Intelligence (AI)
Machine Learning (ML)
Deep Learning (DL)
Supervised Learning
Unsupervised Learning
Neural Networks (Deep Learning)
→ Generative AI (Creates New Content)

Real-World Case Studies

Machine Lerning

Deep Lesrning

Generative AI

NETFLIX

TESLA

Conclusion

- ML, DL, and Generative AI are related but have different purposes.
- DL is a more complex version of ML, while Generative AI focuses on creating new things.
- Understanding these differences helps in using AI effectively.
- AI is advancing quickly, and Generative AI is becoming more powerful.

Thank you for your time!

Have any questions? Feel free to ask!